Why AI Towns Hang Their Own: Herding and Model Lineage in Multi-Agent Social Deduction

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Abstract

Many safety designs trust the combined judgment of several AI agents, which helps only if their views are independent. We test this in Blood on the Clocktower, a social deduction game in which a team of AI players must find a hidden Demon while evil players lie to them. We reanalyze all 538 public games of Clocktower Bench (Patel, 2026), scoring 14,440 daily private forecasts of who the Demon is. First, the good players' beliefs about innocent players are so alike that about eight players score like fewer than three independent forecasters on a redundancy index on day 1 (1.79 to 2.28 by lab) and about one by day 5 (1.09 to 1.15). The index measures belief similarity, not a literal count of independent agents. Second, when a town executes its own top suspect, the execution is almost always right if at least 90% of good players agree (4 wrong of 347), but wrong 63% of the time below that [95% CI 60–67%]. Good players nominated two thirds of the wrong executions, so most of the damage comes from the good team itself. Third, in mixed teams, two copies of the same model agree more than a Claude player and a GPT player (+0.082 correlation [0.065, 0.099], higher in 91% of games), and this holds after we control for private conversation. Agreement among AI agents is weak evidence of independent confirmation, and honest copies of one model can look like a coordinated bloc.

1. Introduction

A common safety pattern is to ask many AI agents the same question: several monitors review one action, several judges grade one answer, or a swarm votes on a plan. The pattern assumes that errors cancel out. They cancel only when the agents fail in different ways. If the agents copy each other, or share a model and therefore share blind spots, a crowd of ten can act like one.

Social deduction games are a clean test bed for this problem. In Blood on the Clocktower, a good team must find a hidden Demon by talking, comparing evidence and voting to execute players, while a small evil team lies and steers the vote. Every day, the good team must turn many private, partly false clues into one group decision. That is the same task a monitoring swarm faces when some of its inputs are adversarial.

Clocktower Bench (Patel, 2026) ran 538 such games between eleven frontier models released in 2026. During each game, every good player was asked, outside the game, for its probability that each other player was the Demon. These belief probes let us measure each agent's forecast directly, without reading it off its speech. We reuse this public data to ask how the town's beliefs form, how independent they are, and when the group decision fails.

Contributions.

- We measure how redundant the members of an AI team are with an effective-number index, check it on simulated agents we know are independent, and show that it falls from about two to about one within the first days of a game.
- We show that only near-unanimous executions are reliable, and that most wrong executions are nominated by good players, not by the evil team.
- We show a lineage effect: in mixed teams, copies of the same model hold more similar beliefs than players of different models, beyond what their private conversations explain.
- We report negative results: standard forecast aggregation methods do not beat a plain average, and the most informed player cannot be picked in advance.

2. Related Work

Multi-agent risks. Hammond et al. (2025) sort failures of AI agent groups into miscoordination, conflict and collusion, and name network effects and destabilising dynamics as risk factors. Our redundancy result is a measured case of a network effect: each agent's belief moves toward the others' until the group adds little independent information.

Collusion detection. Entries in Apart Research's AI Collusion Research Sprint (October 2026) reported that detectors which flag agreement or shared behavior can accuse honest agents that simply run the same model. Our lineage result gives a direct measurement of this effect in a setting where we know who is honest, and shows its size and how it changes over time.

Forecast aggregation. The Brier score (Brier, 1950) is the standard measure of probability forecasts. For human crowds, averaging and then making the average more extreme often improves accuracy (Baron et al., 2014). We test whether these methods transfer to AI crowds. Our effective number of forecasters uses the design-effect formula of Kish (1965).

Correlated errors and conformity in LLMs. Kim et al. (2025) show that across more than 350 LLMs, models make strongly correlated mistakes, even across providers and architectures. Weng et al. (2025) show that LLM agents in groups conform to the majority, and test reflection as a fix. Multi-agent debate (Du et al., 2023) uses several agents to improve answers, which assumes their views add information. Our work differs in three ways: we measure redundancy in a real adversarial game with a known ground truth, we tie it to group decisions that fail (wrong executions), and we separate a same-model effect from private communication in mixed teams.

Social deduction benchmarks. Patel (2026) built Clocktower Bench and reported that newer OpenAI models detect the Demon better, mainly because they weigh each new clue against earlier evidence. A separate Blood on the Clocktower benchmark was posted on LessWrong (Taylor et al., 2026). Patel listed mixed teams and same-model

teammate detection as open questions. We do not rate models. We study the group's beliefs as a forecasting crowd.

3. Methods

Data. We downloaded the game file for every game listed on clocktowerbench.com (538 games: 386 rated, 132 mixed-team, 20 unrated), using the same files the site's replay viewer loads. Each game has 10 to 12 players under protocol v16 or v17, with one model in every good seat and one model in every evil seat. In the 132 mixed games, the good or the evil team is half Claude Opus 5.5 and half GPT-5.6 Sol. Each file holds the seating, true roles, every public message, whisper, nomination and execution, and the belief probes.

Belief probes. Each day, before nominations, each good player is asked for its probability that each other player is the Demon. The probe runs outside the game and does not change it. We normalize each player's probabilities to sum to 1. We use the probe before nominations, or the dawn probe if that is missing.

Scoring. We score each forecast with the multi-class Brier score against the starting Demon (0 is perfect; a uniform guess over all players scores about 0.91). The Demon can pass to another player when it dies, so we stop counting a game on the day the starting Demon dies. This leaves 1,798 game-days.

Agreement and effective number. For each game-day, we compute the Pearson correlation between the belief vectors of every pair of good players over innocent players only: we leave out the real Demon and the two players in the pair. Leaving out the Demon means that two players who both find the right answer do not count as agreeing. For a good player, every belief about an innocent player is an error, so this measures how alike their errors are. The mean of these correlations is the town's agreement r. The effective number is $n / (1 + (n - 1)\, r)$, with r clipped to [0, 1] (Kish, 1965). We use it as a redundancy index, not a literal count of independent agents: it assumes one correlation for all pairs, and it measures how alike beliefs are within a day, not whether errors depend on each other across games.

Control. We checked the measure on 500 simulated towns of 8 agents and 11 players for each of four cases (Table 0). Agents that guess at random, or that each see the truth through their own private noise, score close to 8. Copies of one agent score 1. Agents that add private noise to a shared view of the innocent players, as with shared public evidence, score 2.4. So the measure counts redundant beliefs correctly, but it cannot tell copying apart from shared evidence.

Table 0. Effective number our measure reports for simulated towns of 8 agents.

Simulated agents	Effective number
Independent random guesses	7.14
Truth plus independent private noise	7.23
Shared view plus private noise	2.43
Copies of one agent	1.00

Executions. For each day where the town executed a player, we find the town's top suspect (highest average probability) and the share of good players whose own top suspect is the same player. We keep days where the executed player was the town's top suspect (1,055 executions) and record whether that player was evil and who nominated them.

Lineage. In the 66 mixed games where the good team is mixed and probes exist, we label each pair of good players as same-model or cross-model. We regress pair agreement on a same-model indicator, the log number of whispers between the pair, whether both hold information roles, and day fixed effects, with standard errors clustered by game (8,275 pair-days).

Statistics. Days, executions and player pairs in one game are not independent. So all 95% intervals in the text come from a bootstrap that resamples whole games (2,000 resamples). Table 1 also shows Wilson intervals; the game-level intervals are nearly the same (Appendix B). Regressions use standard errors clustered by game. All analysis uses existing games; we ran no new games.

What did not work. We first measured agreement with the correlation of error vectors. It was above 0.95 everywhere, because every player's error is dominated by the one entry for the real Demon, so it said nothing. Belief correlation without the pair's own entries fixed this. We also tried to find who first named each executed player in chat with a keyword match, but it matched too loosely to trust, so we use nominations instead.

4. Results

4.1 Team members become redundant within two days

About eight good players answer each probe. On day 1 they score 1.79 [1.70, 1.90] on the redundancy index in Claude towns, 1.81 [1.69, 1.94] in GPT towns and 2.28 [2.00, 2.61] in mixed towns. By day 5 these fall to 1.15, 1.12 and 1.09 (Figure 1). Mixed towns start more independent, but the advantage is gone by day 2. Already on day 1, the real towns are less independent than the shared-view control (2.43). By the control in Section 3, this redundancy may come from copying, from shared public evidence, or from both.

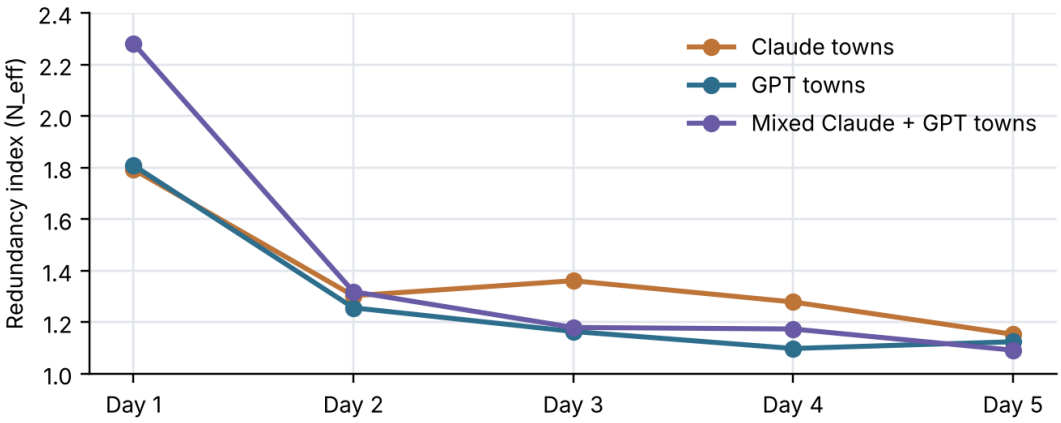


Figure 1. Redundancy index (effective number) among about eight good players, by day. Game-days per day: 537, 483, 410, 322, 189.

Redundancy is not always harmful, because agreement can follow real evidence. In a logistic regression of the good team winning on the town's mean agreement over days 1 and 2, with fixed effects for the good and evil models, more agreement goes with more good wins, but the link is not significant (coefficient 0.22 per standard deviation, p = 0.09, 538 games). The problem is that agreement looks the same whether it comes from shared evidence or from copying.

4.2 Only near-unanimous executions are reliable

Table 1. Executions of the town's top suspect, by the share of good players whose own top suspect was that player.

Town agreement	Executions	Executed a good player	95% CI
Under 75%	364	236 (65%)	60–70%
75 to 90%	344	212 (62%)	56–67%
90% and up	347	4 (1%)	0–3%



Figure 2. Right and wrong executions of the town's top suspect by town agreement. Below 90% agreement, most executions kill a good player.

There is a sharp threshold (Table 1, Figure 2). When at least 90% of good players name the same suspect, the town kills a good player in 4 of 347 executions. Below 90%, it kills a good player in 448 of 708 executions (63% [60, 67]). Players sometimes give two suspects the same top probability (5.8% of player-days). With tie-aware agreement, where a tied player gives each tied suspect partial credit, the result holds: 63% wrong below 90% and 0.3% at 90% and up. A town that is "mostly sure" is wrong more often than it is right. Wrong executions cost the game: in the 175 games with a wrong execution at 75% agreement or more, the good team won 40% [33, 47], against 68% across all games.

Who starts it. An evil player nominated 32% [28, 37] of wrong executions and 6% [4, 8] of right ones (Fisher's exact $p < 10^{-28}$). Evil holds about 27% of seats, so evil players do push the town toward innocent players. But good players nominated the other 68% of wrong executions. Most of the damage comes from the town itself.

4.3 Copies of a model agree more than different models

In mixed good teams, same-model pairs hold more similar beliefs than cross-model pairs (Figure 3). Per game, the mean gap is +0.082 [0.065, 0.099], and the gap is positive in 91% of the 66 games. In the regression, being the same model adds 0.078 to pair agreement ($p < 0.001$), while more private whispers add 0.043 per log-unit ($p = 0.001$) and both holding information roles adds nothing ($p = 0.96$). Same-model pairs whisper only a little more than cross-model pairs (7.5 against 6.7 whispers per game). The gap is largest on day 1 (0.149) and shrinks each day to 0.044 on day 5, but it stays positive when day 1 is left out (+0.060 [0.044, 0.078]).

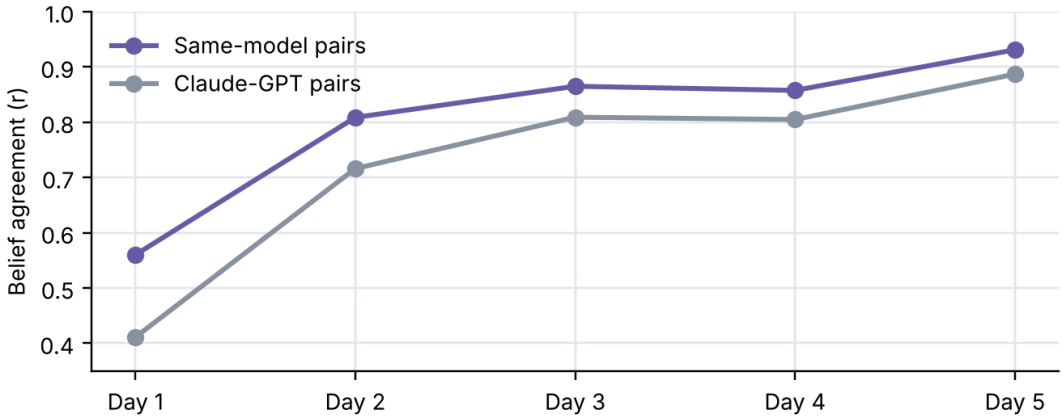


Figure 3. Mean belief agreement between pairs of good players in mixed Claude Opus 5.5 and GPT-5.6 Sol teams, by day.

Interpretation. Both kinds of pair see the same public evidence, so shared evidence cannot explain the gap. We read the gap as each model's built-in priors about who looks suspicious, which copies share and which discussion slowly washes out. Patel (2026) noted that no player ever used writing style to find teammates. Our result shows that copies align without needing to find each other. The first probe of each game comes

after the first day's discussion, so we cannot say the gap exists before any talk; we can say it is largest on day 1 and is not explained by private whispers.

4.4 Negative results

- **Aggregation.** The plain average of the town scores 0.866 Brier, against 0.908 for a uniform guess. The median (0.898), a log pool (0.879) and confidence weighting (0.884) do not beat it. Extremizing, which helps human crowds (Baron et al., 2014), makes it much worse (0.980).
- **Picking the best player.** On 95% of game-days, at least one good player beats the town average. But players with information roles score only a little better than others (0.899 against 0.917), and players far from the crowd are no better (0.910 against 0.908). We found no way to pick the best player in advance.
- **Calibration.** Both labs are overconfident at the top. When a Claude player gives 75–100%, the player is the Demon 39% of the time. For GPT players, the figure is 64%.

5. Discussion and Limitations

Implications for safety. Groups of AI monitors, judges or voters are often trusted because they agree. Our results suggest three rules for such designs. First, measure how redundant the judges are, not just the head count. In our data, a panel of eight scored below two on the redundancy index. Second, treat partial agreement as weak evidence. In our data, only near-unanimity was reliable. Third, do not use agreement alone as a sign of collusion. Honest copies of one model agree more than honest agents of different models, so an agreement-based collusion detector will flag model monocultures. Mixing models raises early independence, but in our data this effect was gone by day 2.

Limitations. All results are observational and come from one game, one script (Trouble Brewing) and one mixed pair of models. Probes are self-reports given outside the game, and may differ from what a player acts on. Later days have fewer games and lean toward games evil is winning. Our stopping rule removes days after the starting Demon dies. The agreement measure cannot tell copying apart from shared public evidence (Table 0), and does not show who copied whom. Appendix A discusses these further.

Future work. The cheapest next step is an intervention: require each good player to state the strongest case against the top suspect before each vote, and test whether this lowers wrong executions below 90% agreement. Other steps are to repeat the lineage test with more model pairs, and to test whether evil teams learn to exploit the agreement threshold.

6. Conclusion

In 538 games of AI social deduction, good teams became highly redundant within days, scoring about one on the redundancy index, hung an innocent player in most executions

made without near-unanimity, and did most of that harm themselves. Copies of one model agreed with each other more than with other models, beyond what their private conversation explains. For multi-agent oversight, a crowd of AI agents should be counted by its independent views, not its size.

7. Code and Data

Data: the public game files from clocktowerbench.com, produced by Clocktower Bench (Patel, 2026; code at github.com/dwarkeshsp/clocktower). The game engine and all games are pre-existing third-party work; we ran no games. All analysis code (Python: data extraction, scoring, statistics and figures) was written during the sprint. One script, run_all.sh, downloads the public data and reproduces every number and figure in this report in about five minutes, with no API keys. The repository is linked from the submission form. There is no information hazard: the work uses public game logs and reports aggregate statistics.

8. LLM Usage Statement

Claude (Anthropic) was used to write and run the analysis code, compute the statistics and draft the text. Every number in this report comes from script output on the downloaded data. During the work, the analysis was revised three times after checks showed problems: the error-correlation measure (Section 3), the claim that medium agreement is uniquely dangerous, which the confidence intervals did not support, and, after a separate AI-assisted audit, the agreement measure, the statistics and the tie handling (Appendix B). The author reviewed the report and is responsible for its content. References were checked against their sources.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Limitations

- **Observational data.** We ran no experiments. The links we report between agreement, nominations and outcomes are correlations. The lineage gap is measured in the same games and the same public evidence for both pair types, which rules out shared evidence as its cause, but not every other cause.
- **One game, one script, one model pair.** All games use the Trouble Brewing script. The lineage result rests on one pair of models (Claude Opus 5.5 and GPT-5.6 Sol) in 66 games. Other pairs, other games and real monitoring tasks may differ.
- **Probes are self-reports.** Probes are asked outside the game. A player's stated probabilities may differ from the beliefs it acts on, and the models may be poorly calibrated when reporting them (Section 4.4).
- **Selection over days.** Long games are more often games evil is winning. Late-day numbers come from fewer and less typical games.
- **Stopping rule.** We stop counting each game when the starting Demon dies, because the Demon can pass to another player. Days after this point are not in our analysis.
- **Measures.** Our agreement measure is a mean pairwise correlation over innocent players, and the effective number formula assumes equal correlation between all pairs. Our simulation control shows the measure scores independent agents as independent, but agents sharing the same public evidence also score as redundant (Table 0). So the measure shows redundancy, not copying. The lineage result does not depend on this, because same-model and cross-model pairs see the same public evidence. Our nomination analysis covers only executions of the town's top suspect.
- **Multiple analyses.** We ran several analyses on the same data, and some were chosen after early looks. We report the negative results we found, but the reported p-values are not corrected for this.

Dual-use and ethical considerations

The work uses public logs of games played by AI agents. No people took part and no personal data is used. The main dual-use risk is that knowing how AI groups herd could help someone build agents that steer a group of AI monitors, for example by pushing a target until the group passes a voting threshold. We judge this risk low: the effect is visible in public data, and the findings mainly help defenders measure and correct herding. The game requires deception by evil players. We report how deception affects group decisions and do not provide methods to make agents more deceptive.

Appendix B. External audit and robustness checks

An AI-assisted audit of an earlier draft raised the concerns below. We tested each one. The code for every check is in the repository (analysis/02_independence.py and analysis/07_robustness.py).

Table 2. Audit concerns, checks and outcomes.

Concern	Check	Outcome
Agents may agree only because they found the same right answer.	Agreement now uses beliefs about innocent players only, leaving out the real Demon.	Effective number on day 1 moved from 1.72–2.17 to 1.79–2.28. Findings unchanged.
The formula may score independent agents as one.	Simulated towns with known structure (Table 0).	Independent agents score 7.1–7.2 of 8; copies score 1.0. Shared evidence also scores as redundant (2.4), so we now say "redundant", not "copying".
Related results counted as separate observations.	Bootstrap over whole games; regressions clustered by game.	Wrong rate below 90%: 63% [60, 67], same as Wilson. Evil-nominated wrong executions: 32% [28, 37]. Lineage gap: per-game mean +0.082 [0.065, 0.099], the estimate used in the text; pooling all pairs and resampling games gives +0.083 [0.066, 0.103].
Ties in a player's top suspect.	Tie-aware agreement with partial credit.	5.8% of player-days tie. Below 90%: 63% wrong; 90% and up: 0.3% wrong.
Novelty against prior work on debate, correlated errors and conformity.	Added Kim et al. (2025), Weng et al. (2025) and Du et al. (2023) to Related Work.	Contribution now framed as measurement in an adversarial game with ground truth, plus the lineage result.
Claims that change with the corrected measure.	Reran all analyses from a clean download.	Early agreement and winning: no longer significant ($p = 0.09$). Best player beats the town average on 95% of days, not 98%.